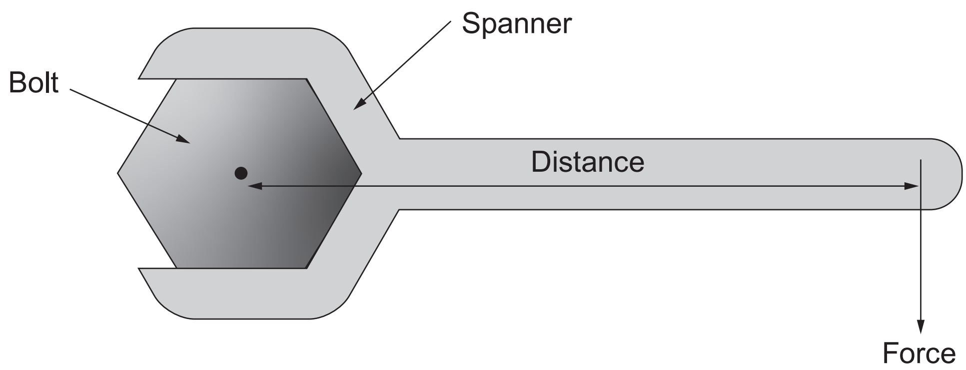
Examiner only

3. The diagram shows a spanner being used to tighten a bolt.



A force at the end of the spanner produces a moment about the bolt.

(a) In the diagram above, the force is 80 N and the distance is 0·25 m.

Use the equation:

$$\text{moment} = \text{force} \times \text{distance}$$

to calculate the moment of the force about the bolt. [2]

Moment = ... N m

(b) Henry says it is easier to tighten the bolt if a longer spanner is used.
Explain why Henry is correct. [2]

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